Touchline

Teaching AI football players to follow a learned manager's tactics
Project scope for the team • 6 Nov 2026

> **In one paragraph.** We train a full football team with reinforcement learning (RL) in a football video game made for AI research. Eleven AI players learn to play together. A separate AI "manager" watches the match and switches tactics as it goes (press high, sit deep, play a false nine). It talks to the players through simple instructions such as *press*, *mark* or *hold the line*, and the players learn to follow them while still trying to win. We want to show that a manager who adapts mid-match beats any single fixed tactic, and that the players really change how they play when told to.

This document explains everything you need to follow the project, assuming you don't know the game, football tactics or the RL methods. Section 7 has a glossary.

1. The game: Google Research Football

Google Research Football (GRF) is a free football video game built by Google so that AI can learn to play it. It plays real 11-a-side matches with proper rules (fouls, offside, cards, set pieces). Every few milliseconds the game tells each AI player what is happening and asks what it wants to do. Its built-in computer opponent comes in easy, medium and hard, and it is the standard benchmark: published AI papers report how well their agents do against it.



Our modified game: Barcelona (blue and garnet stripes) against Real Madrid (white), both teams played by the built-in AI.

What we have already changed

- **Realistic physics** for the ball and players, and **kick power** the AI controls (how hard to pass or shoot), not just which kick.
- **19 player types (archetypes)**, for example target man, false nine, ball-winning midfielder. They all have the same total stat points, so none is simply better; you trade pace for strength and so on. A player's type can be changed during a match.
- **Full 90-minute matches** with injury time, team and player names, kits, replays.
- **Human play** with a PlayStation controller, with on-pitch markers, for demos and for testing our AI against a real person.
- **135 automated tests**, so we know changes don't break anything.

> **Important:** all the human-play extras (markers, kits, names, controller assists) are switched off by default. Training sees the normal game.

2. The football you need to know

Positions

Position	Short	Job in one line
Goalkeeper	GK	Stops shots; the only player allowed to use hands.
Centre-back	CB	Central defender; stops attackers, wins headers.
Full-back	LB / RB	Defender on the side; also runs forward down the wing.
Defensive midfielder	DM	Sits in front of the defence, breaks up attacks.
Central midfielder	CM	Links defence and attack; passes, runs, tackles.
Winger	LM / RM	Attacker on the side; dribbles, crosses, cuts inside.
Striker	ST / CF	Main goal scorer, furthest forward.

Tactics we use

A tactic is a plan for how the whole team plays. Our manager picks between six:

Tactic	What it means
High press	Chase the ball high up the pitch to win it back near their goal.
Low block	Drop deep, stay compact, give them the ball but no space.
Possession	Keep the ball with short passes; patient build-up.
Counter	Defend deep, then break forward fast as soon as we win the ball.
Wing play	Attack down the sides and cross into the box.
False nine	The striker drops into midfield to pull defenders out; wingers run into the space behind.

Offside, briefly

An attacker is offside if, when a teammate passes to him, he is nearer the opponents' goal than both the ball and the second-last defender. It stops players waiting next to the goal. The game enforces it automatically; it matters to us because good runs have to stay onside.

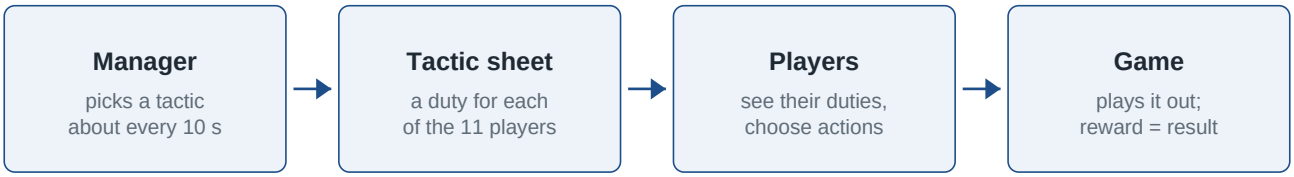
3. The RL you need to know

Term	Plain meaning	In our project
Agent	Something that learns to act by trial and error.	Each player, and the manager.
Observation	What the agent sees each step, as a list of numbers.	Positions, ball, score... 957 numbers per player.
Action	What the agent does each step.	Run direction, speed, kick type, tackle, kick power.
Reward	A score that tells the agent how it did.	Goals, winning, and early on a bonus for following instructions.
Policy / network	The neural network that picks actions.	One network shared by all 11 players; one for the manager.

Term	Plain meaning	In our project
PPO	A standard, reliable RL algorithm for training a policy.	How both players and manager learn.
MAPPO	PPO for many agents: shared network, plus a 'critic' that sees the whole match while training to judge how good things are.	How the 11 players learn together.
Self-play	Training against copies of yourself.	Mixed with older versions and the built-in AI so it doesn't go in circles.
Curriculum	Start with easy tasks, then harder ones.	Drills, then 5v5, then full matches.

4. How it fits together

The game has no "tactics button". So a tactic reaches the players as instructions we call **duties**, which are part of what each player sees. This is how one decision travels:



Duties

A duty is one simple job for one player. Every player always has three: one for when they have the ball, one for when we have it, and one for the first five seconds after the ball changes hands. A tactic is simply a duty for every position plus three team settings (how high the defence sits, how eagerly to press, how wide to play). These are the duties in the first version:

When	Duty	What the player does
They have the ball	Hold zone	Stay near your spot in the team shape.
	Press	Close down the player with the ball and try to win it.
	Cover	Stay 5-10 m behind a pressing teammate in case he is beaten.
	Mark	Stay close to one opponent, between him and our goal.
	Hold line	Stay level with the other defenders (the offside line).
We have the ball	Hold shape	Stay near your spot, shifted forward.
	Support	Be 8-15 m from the ball carrier with a clear passing line.
	Width	Stay near the touchline on your side.
	Run in behind	Get beyond the last defender, onside, toward goal.
Just lost the ball	Recover	Sprint back toward your defensive spot.

How players learn to follow duties

- A **duty check** answers "is this player doing his duty right now?".
- Early in training, players get a small reward whenever their check is true. This bonus is turned down to zero later, so in the end they are rewarded only for winning.
- During training, duties are assigned at random and change every 30-60 seconds, so players learn to follow *any* instruction, not one tactic.
- The same checks later prove the tactics work, e.g. the false nine really plays deeper.

5. What is in and out of scope

Manager ability	Status	How
Change tactics mid-match	In	Through duties. Python work (tickets T10, T20); no game-engine change needed.
Change a player's role (e.g. CM to false nine)	In	Same thing: new duties and a new home position for that player.
Change a player's type (e.g. to target man)	Built, later version	Already in the game; the manager uses it in a later version.
Set plays (corners, free kicks)	Later version	Duties given at restarts; needs extra drills.
Real substitutions (bench player, fresh legs)	Out	The game has no bench. Would need an engine change.
Control the built-in AI's tactics	Out	It only has a difficulty level. We use our own team with random tactics as varied opponents instead.

Also out of scope for this month: copying a human or the built-in AI as a starting point (imitation learning), image-based observations, and anything needing a GPU. These are possible later improvements, not needed for the result.

How we measure success

Question	Measure
Can our players play?	Win rate against the built-in AI (easy, medium, hard).
Do they follow duties?	Share of time each duty check is true.
Do tactics change play?	E.g. false nine plays 8 m deeper; high press wins the ball higher up.
Does the manager help?	Goal difference over 100 matches vs the best fixed tactic.

6. The plan

Week	What happens
1 (10-16 Nov)	Build the pieces: training code, what players see, rewards, drills, results script, cloud machine.
2 (17-23 Nov)	Players learn duties in drills, then 5v5. Manager practises with a rule-based team. **Fri 21 Nov: decide 11v11 or 5v5.**
3 (24-30 Nov)	Full training; manager retrained with the real players; results charts and clips.
4 (1-8 Dec)	No new features. Final numbers, demo, slides, practice.

Who does what

Person	Owns
P1 Player training	Training the 11 players together (MAPPO); runs the training stages.
P2 Env engineer	Everything between game and training: what players see, actions, duty checks, rewards.
P3 Training and testing	Drills, the rule-based team, nightly results, charts.
P4 Manager + PPO	The learning algorithm, the six tactic sheets, the manager.
P5 Ops and presentation	Cloud machine, demo, clips, slides.

> **If things go slowly.** The main risk is computing power: the game runs only on the CPU and full 11v11 training is slow. We have a result at every level: **full goal** 11v11 players plus manager; **if short on compute** the same at 5v5; **guaranteed minimum** a manager trained on a rule-based team plus players trained on drills.

Every task is a ticket with an owner, a due date and a test that says when it is done. The full list, tracker and colour-coded calendar are in the shared tickets doc.

7. Glossary

Word	Meaning
Archetype	A player type with its own mix of stats, e.g. target man (strong, good in the air, slow).
Built-in AI	The game's own computer opponent: easy, medium, hard.
Checkpoint / model	A saved copy of a trained network.
Critic	The part of MAPPO that judges how good a situation is, to guide learning.
Drill	A small practice game, e.g. 2v2, to learn one duty.
Duty	One simple job for one player, e.g. press or mark.
Engine	The game's core code (C++). We changed it for physics, kits, etc.
GRF	Google Research Football, the game we use.
Manager	Our AI that picks the team's tactic during the match.
Tactic / preset	A full set of duties for all 11 players, e.g. high press.
v1	The first version: fewer duties and only the six tactics.